Data Structures and Algorithms · 21CSC201J

Unit 2 · Session 10 · SLO 2 — Solved Worksheet

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Unit â€“ 2 : Session â€“ 10 : SLO â€“ 2 | Doubly Linked List â€” Insertion at the Middle

I. Insert a New Node VAL After a Given Node NUM

- Step 1: IF AVAIL = NULL
- Write OVERFLOW
- Go to Step 12
- [END OF IF]
- Step 2: SET NEW_NODE = AVAIL
- Step 3: SET AVAIL = AVAIL -> NEXT
- Step 4: SET NEW_NODE -> DATA = VAL
- Step 5: SET PTR = START
- Step 6: Repeat Step 7 while PTR -> DATA != NUM
- Step 7: SET PTR = PTR -> NEXT
- [END OF LOOP]
- Step 8: SET NEW_NODE -> NEXT = PTR -> NEXT
- Step 9: SET NEW_NODE -> PREV = PTR
- Step 10: SET PTR -> NEXT -> PREV = NEW_NODE
- Step 11: SET PTR -> NEXT = NEW_NODE
- Step 12: EXIT

II. Insert a New Node VAL Before a Given Node NUM

- Step 1: IF AVAIL = NULL
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- [END OF IF]
- Step 2: SET NEW_NODE = AVAIL
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- [END OF LOOP]
- Step 8: SET NEW_NODE -> NEXT = PTR
- Step 9: SET NEW_NODE -> PREV = PTR -> PREV
- Step 10: SET PTR -> PREV -> NEXT = NEW_NODE
- Step 11: SET PTR -> PREV = NEW_NODE
- Step 12: EXIT